

# ONUR KELEŞ, Ph.D.

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## Research Interests

New model/learning paradigms · Non-linear neuron models · Image and video super-resolution · Learned image compression · Generative video synthesis (talking-head generation, lip synchronisation) · Gaussian splatting

## Education

### Ph.D. | *Image Processing with Deep Learning, Electrical & Electronics Engineering*

Sep. 2019 – Mar. 2025

Koç University

İstanbul, TÜRKİYE

- Graduation GPA: 4.00/4.00
- Dissertation: Non-linear Neuron Modeling using Padé Approximants with Applications to Single Image Super-Resolution and Image Compression
- Advisor: Prof. A. Murat Tekalp

### M.Sc. | *Signal Processing, Electrical & Electronics Engineering*

Sep. 2016 – Aug. 2019

Boğaziçi University

İstanbul, TÜRKİYE

- Graduation GPA: 3.50/4.00
- Thesis: Vision Correction for the Visually Impaired via Digital Image Processing

### B.Sc. | *Signal Processing, Electrical & Electronics Engineering*

Sep. 2010 – June 2016

Boğaziçi University

İstanbul, TÜRKİYE

- Graduation GPA: 3.24/4.00
- Design Project: Automatic Punctuation Generation for Turkish

### Exchange Program | *Chinese Language and Culture*

Feb. 2014 – July 2014

Shanghai University

Shanghai, CHN

## Publications

- Onur Keleş**, A. Murat Tekalp, "Padé Neurons for Efficient Neural Models," in *IEEE Transactions on Image Processing*, vol. 35, pp. 1508–1520, 2026, DOI: 10.1109/TIP.2026.3653202.
- Onur Keleş**, A. Murat Tekalp, "Paon: A New Neuron Model Using Padé Approximants," in *IEEE International Conference on Image Processing (ICIP)*, 2024, pp. 207–213, DOI: 10.1109/ICIP51287.2024.10648214 (<https://github.com/onur-keles/Paon>).
- Onur Keleş**, A. Murat Tekalp, Junaid Malik, and Serkan Kiranyaz, "Self-Organized Residual Blocks for Image Super-Resolution," in *IEEE International Conference on Image Processing (ICIP)*, 2021, pp. 589–593, DOI: 10.1109/ICIP42928.2021.9506260.
- M. Akın Yılmaz, **Onur Keleş**, Hilal Güven, A. Murat Tekalp, Junaid Malik, and Serkan Kiranyaz, "Self-Organized Variational Autoencoders (Self-VAE) for Learned Image Compression," in *IEEE International Conference on Image Processing (ICIP)*, 2021, pp. 3732–3736, DOI: 10.1109/ICIP42928.2021.9506041.
- Onur Keleş**, M. Akın Yılmaz, A. Murat Tekalp, Cansu Korkmaz, and Zafer Doğan, "On the Computation of PSNR for a Set of Images or Video," in *Picture Coding Symposium (PCS)*, 2021, pp. 1–5, DOI: 10.1109/PCS50896.2021.9477470.
- Onur Keleş**, and Emin Anarım, "Adjustment of Digital Screens to Compensate the Eye Refractive Errors via Deconvolution," in *Ninth International Conference on Image Processing Theory, Tools and Applications (IPTA)*. IEEE, 2019, pp. 1–6, DOI: 10.1109/IPTA.2019.8936098.

## Work Experience

### AI Research Scientist ⇒ Senior AI Research Scientist

Oct. 2022 – June 2026

Codeway Digital Services

İstanbul, TÜRKİYE

- Doubled the output resolution of a top-chart mobile application for face image enhancement, improving image clarity. Contributed to product: *Pixelup*.
- Developed a fast, efficient solution for multi-face recognition and separation for identity-specific face generation. Contributed to products: *Pixelup*, *Retake*.
- In a team, improved the naturalness of a face reenactment model outputs. Contributed to product: *FaceDance*.
- Led a 4-person team to develop and improve lip synchronization, video dubbing, and talking head generation solutions.
- Increased head video generation from 90 FPS to 2800 FPS with enhanced fidelity using Gaussian splatting, providing the ability for real-time deployment.

### Doctoral Researcher

Sep. 2019 – Mar. 2025

Koç University İş Bankası Artificial Intelligence (KUIS AI) Center

İstanbul, TÜRKİYE

- Advanced research on non-linear neuron models in super-resolution and image compression, published three peer-reviewed conference papers detailing research outcomes.
- Proposed a novel, inherently non-linear neuron model, applied the model to super-resolution and image compression problems, resulting in a published conference paper and a journal paper.

### Software Specialist (Part-time)

Feb. 2016 – June 2016

Speech-Enabled Software Technologies (SESTEK)

İstanbul, TÜRKİYE

- Initiated development of text-to-speech (TTS) software, and prepared a dictionary, for Chinese language.
- Enhanced automatic punctuation generation for Turkish text produced by speech-to-text (STT) software.

## Research Projects

- The Scientific and Technological Research Council of Türkiye (TÜBİTAK) Project** Spr. 2021 – Spr. 2024  
Koç University İş Bankası Artificial Intelligence (KUIS AI) Center, Koç University
- Project: Explainable Deep Learning Approaches for Image and Video Restoration and Compression (120C156)
- The Scientific and Technological Research Council of Türkiye (TÜBİTAK) Project** Fall 2019 – Spr. 2021  
Koç University İş Bankası Artificial Intelligence (KUIS AI) Center, Koç University
- Project: Image and Video Processing with Deep Learning (217E033)
- Boğaziçi University Scientific Research Project** Spr. 2019  
Boğaziçi University
- Project: Adjustment of Digital Screen to Compensate the Eye Refractive Errors via Image Processing (19A02P1)

## Teaching Experience

*Prepared problem sets; prepared and graded quizzes and MATLAB coding homeworks.*

- ELEC201 – Signals and Systems | Teaching Assistant** Spr. 2022  
Koç University
- ELEC301 – Systems, Control and Communications | Teaching Assistant** Fall 2019 – 2021  
Koç University
- ELEC303 – Digital Signal Processing | Teaching Assistant** Spr. 2020 – 2021  
Koç University
- ELEC406/506 – Digital Image and Video Processing | Teaching Assistant** Spr. 2021; Fall 2022, 2024  
Koç University

## Professional Service

- Reviewer | IEEE International Conference on Image Processing (ICIP)** 2022 – 2024

## Conferences and Presentations

### Conferences

- IEEE International Conference on Image Processing, Oct. 2024, Abu Dhabi, UAE
- IEEE International Conference on Image Processing, Sep. 2021, Anchorage, Alaska, USA (*online*)
- Picture Coding Symposium, June 2021, Bristol, GBR (*online*)
- International Conference on Image Processing Theory, Tools & Applications, Nov. 2019, İstanbul, TÜR

### Invited Talks

- Self-Organized Operational Neural Networks in Image Super-Resolution*, KUIS AI Meetings, Feb. 2021, İstanbul, TÜR
- Endüstride Yapay Zeka Uygulamaları*, BAYÖYO TEDÜ-ODTÜ, July 2024, Ankara, TÜR

## Achievements, Honors and Awards

- TÜBİTAK Project Scholarship** Fall 2019 – Spr.2024  
Koç University
- Boğaziçi University Scientific Research Project Scholarship** Spr. 2019  
Boğaziçi University
- TÜBİTAK 2210-A National Grade Scholarship** Fall 2016 – Fall 2018  
Boğaziçi University
- Academic Personnel and Graduate Education Entrance Exam (ALES)** Sep. 2018  
National graduate admissions exam; ranked 41<sup>st</sup> out of around 120,000 participants in Türkiye
- University Placement Exam (YGS & LYS)** June 2010  
National university placement exam; ranked 485<sup>th</sup> out of 1.8 million participants in Türkiye
- Turkish National Folk Dance Competition** 2003  
1<sup>st</sup> place in the nation-wide contest

## Skills and Abilities

- Programming:** Python, MATLAB, C/C++, C#
- Frameworks:** PyTorch, NumPy
- Languages:** Turkish (native), English (proficient), Chinese (pre-intermediate)